Explainable Artificial Intelligence

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***Abstract*—With the increasing use of machine learning (ML) and deep learning (DL) in critical areas such as healthcare and finance, the opacity of model decision-making is a growing concern. Explainable Artificial Intelligence (XAI) aims to develop methods and tools that make the decisions of learning models transparent and interpretable to humans. This paper reviews the motivation, concepts, taxonomy, and techniques used in XAI, including intrinsically interpretable models and post-hoc explanation methods such as Local Interpretable Model-agnostic Explanations (LIME), SHapley Additive exPlanations (SHAP), and Gradient-weighted Class Activation Mapping (Grad-CAM). It presents the mathematical foundations of the major methods, compares them qualitatively on image data, and discusses the trade-off between model accuracy and interpretability, application domains, evaluation of explanations, regulatory and ethical dimensions, and challenges in developing explanation systems that are faithful, stable, and understandable to non-experts. Finally, the paper outlines open research directions in explainable AI.**

***Index Terms*—Explainable Artificial Intelligence, Interpretability, Machine Learning, Black-Box Models, LIME, SHAP, Grad-CAM, Model Transparency, Trustworthy AI**

I. Introduction

Artificial Intelligence (AI) systems based on deep learning have achieved remarkable results in image recognition, natural language processing, and medical diagnosis [12]–[14]. However, their complexity makes decision-making difficult to understand, giving rise to the *black-box problem*. This lack of transparency raises concerns about accountability, fairness, and safety, and hinders adoption in critical domains where human oversight is necessary.

Explainable Artificial Intelligence (XAI) develops techniques that make the behavior of AI models understandable to humans while maintaining predictive performance. The goal is to explain not only *what* a model predicts but *why* [7], [8]. Interest has grown because models now influence credit, medical, hiring, and sentencing decisions; because models can achieve high accuracy for the wrong reasons, such as spurious background cues [1] or biased proxy variables [15]; and because regulators increasingly expect automated decisions to be documented and contestable [16], [17].

This paper gives a structured overview for readers familiar with basic machine learning. Section II defines key concepts, Section III explains the need for explainability, Section IV reviews related work, and Section V presents a taxonomy. Sections VI and VII cover interpretable models and post-hoc methods, Section VIII compares methods, and Section IX discusses evaluation. Sections X–XII address applications, regulation and ethics, and challenges, and Sections XIII and XIV give future directions and conclusions.

II. Background and Key Concepts

A. Interpretability Versus Explainability

Interpretability is commonly described as the degree to which a human can understand the cause of a decision [6], [10]. Explainability usually refers to producing additional information, such as a heatmap or ranked feature list, that clarifies a model's behavior. An interpretable model is understandable by construction, whereas an explainable system may apply a separate explanation mechanism to an opaque model. Lipton cautions that interpretability is not a single well-defined property and that claims should state the desideratum served, such as trust, causality, or fairness [4].

B. Formal Setting

Let $f : \mathcal{X} \to \mathbb{R}$ be a trained model mapping an input $\mathbf{x} = (x_1, \ldots, x_d)$ to a score, such as a class probability. The most common explanation is a vector of attribution scores $\boldsymbol{\phi}$ in which ϕ_i quantifies the contribution of feature i. An *additive* explanation has the form

$$f(\mathbf{x}) \approx \phi_0 + \sum_{i=1}^{d} \phi_i, \tag{1}$$

where ϕ_0 is a baseline, typically the average prediction. LIME and SHAP can both be viewed as ways of choosing these coefficients.

C. Desirable Properties of Explanations

Good explanations are commonly characterized by *fidelity* (faithfulness to the model), *comprehensibility* to the intended audience, *stability* for similar inputs, *sparsity* (few factors), and *actionability* (what could be changed to alter the outcome). These properties often conflict; for example, greater sparsity usually reduces fidelity (Section XII).

III. Need for Explainability

A. Trust

Users are more likely to accept AI systems when decisions can be understood and verified. A loan system can give reasons for approval or rejection, helping applicants and institutions understand its output. Good explanations help users *calibrate* reliance, trusting the system where its reasoning is sound and questioning it where it is not.

B. Accountability

In regulated domains such as banking and healthcare, organizations may need to justify automated decisions. Explanations help decision-makers understand the factors behind an outcome and provide a record that auditors and oversight bodies can review.

C. Debugging and Improvement

Explanations help developers find errors, unintended correlations, and biases. A classifier may separate two animal classes by relying on snow in the background [1]; such *shortcut learning* yields high test accuracy that does not transfer. Inspecting explanations can reveal the problem before deployment.

D. Regulatory Compliance

Governance frameworks increasingly emphasize transparency. Explainability can help organizations document and communicate the factors behind automated decisions. The scope of any "right to explanation" has been debated [16], [17], and legal requirements depend on the regulation and context (Section XI).

E. Fairness

XAI can reveal whether a model relies on sensitive features or their proxies. A widely cited study found that a healthcare risk algorithm using health costs as a proxy for need gave systematically lower scores to Black patients at the same level of illness [15]. Explanations do not establish fairness, but they give auditors evidence to examine [18].

IV. Related Work

Early interpretable models, such as linear regression and decision trees, are transparent because their logic can be inspected. As ensembles [19] and deep networks became common, researchers developed post-hoc techniques that explain predictions without changing the model. Breiman's "two cultures" essay anticipated this debate between interpretable data models and accurate algorithmic models [20].

Attribution and surrogate methods. Ribeiro *et al.* introduced LIME, which approximates a model locally with an interpretable surrogate [1], and later Anchors, which give high-precision if-then rules [21]. Lundberg and Lee proposed SHAP, based on the Shapley value from cooperative game theory [2], [22], and later gave a polynomial-time algorithm for tree ensembles [23]. DeepLIFT propagates contributions relative to a reference activation [24].

Gradient-based and visual methods. Selvaraju *et al.* introduced Grad-CAM, which highlights image regions relevant to a class [3]; Grad-CAM++ improves localization of multiple instances [30]. Earlier work includes gradient saliency maps [25], deconvolution [26], guided backpropagation [27], layer-wise relevance propagation [28], and integrated gradients [29].

Concept, example, and effect-based methods. TCAV measures sensitivity to human-defined concepts [31], influence functions identify influential training points [32], and counterfactuals describe the smallest change that alters a decision [17]. Partial dependence [33], ICE [34], and ALE plots [35] show feature effects.

Surveys and critiques. Guidotti *et al.* [5], Adadi and Berrada [7], Arrieta *et al.* [8], and Samek *et al.* [9] survey the field, and Miller draws on the social sciences [10]. Rudin argues that high-stakes decisions should use inherently interpretable models [11]. Critical work shows that some saliency maps are insensitive to learned parameters [36], that interpretations can be fragile [37], and that LIME and SHAP can be manipulated [38].

V. Taxonomy of XAI Methods

XAI methods can be organized along several axes. *Intrinsic* methods restrict the model so it is understandable by design, while *post-hoc* methods are applied after training. *Local* explanations describe one prediction, *global* ones describe overall behavior. *Model-agnostic* methods (LIME, KernelSHAP, PDP) only query the model, whereas *model-specific* methods (Grad-CAM, TreeSHAP) use its structure and are faster but less portable.

Fig. 1 summarizes a taxonomy by explanation modality.

Heatmap-based methods assign importance to pixels, regions, or tokens (Grad-CAM, saliency maps, LRP, integrated gradients). They are intuitive but easily over-interpreted (Section VII-E). **Concept-based** methods such as TCAV test whether a human concept, like "striped" for "zebra," influences a prediction [31]. **Feature-based** methods (SHAP, LIME, Gini importance, decision trees) report per-variable scores or rules. **One-feature-based** methods (PDP, ICE, ALE) show how the output varies with a single feature. **Fuzzy-logic-based** models

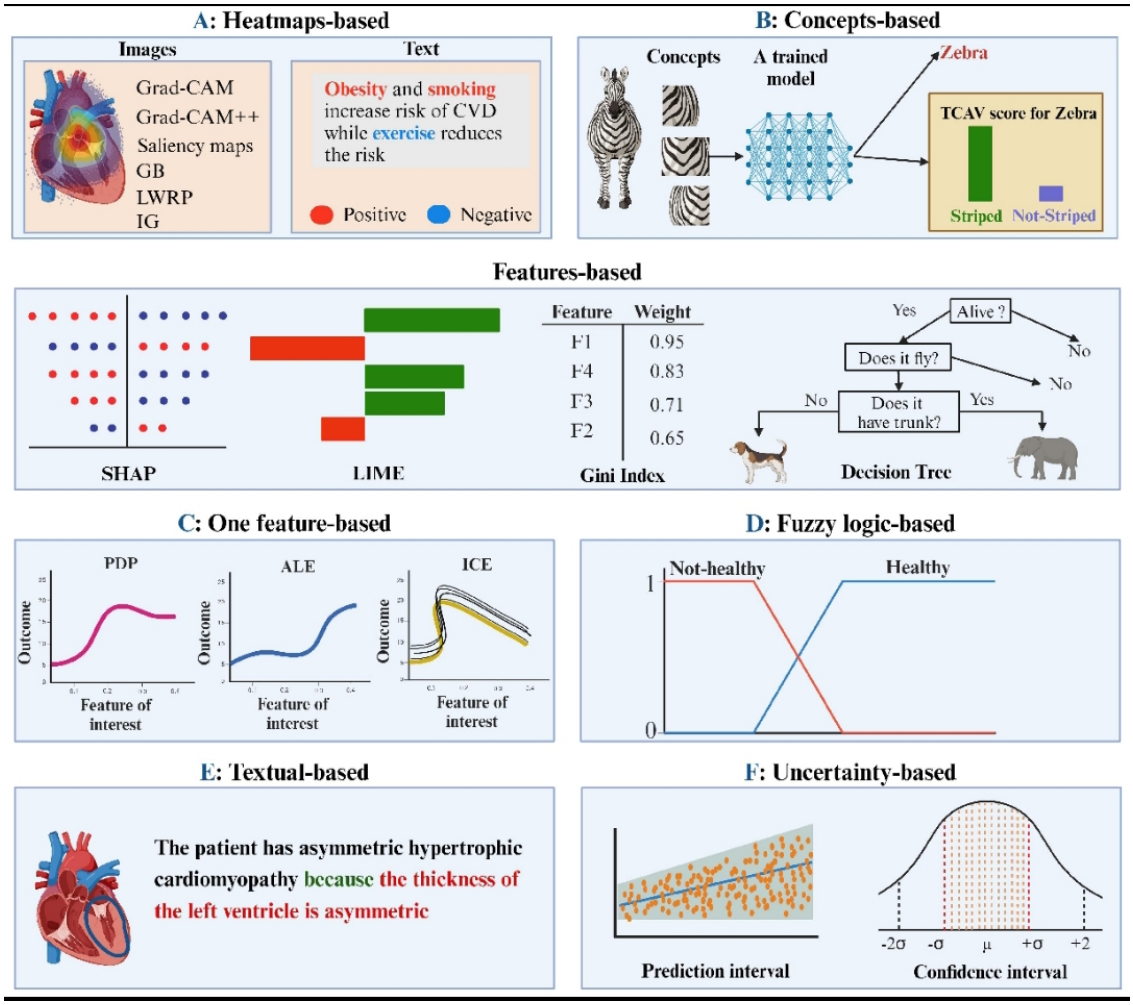


Fig. 1. Categories of XAI methods by explanation modality: (A) heatmap-based (e.g., Grad-CAM, Grad-CAM++, saliency maps, guided backpropagation, LRP, integrated gradients), (B) concept-based (e.g., TCAV), features-based (e.g., SHAP, LIME, Gini importance, decision trees), (C) one-feature-based (PDP, ALE, ICE), (D) fuzzy-logic-based, (E) textual, and (F) uncertainty-based explanations. Adapted from the literature; source to be cited.

use linguistic variables and membership functions [40], giving rules close to natural language. **Textual** explanations give natural-language justifications, which are accessible but may sound plausible without being faithful. **Uncertainty-based** methods communicate confidence through prediction intervals or calibrated probabilities, using techniques such as Monte Carlo dropout [41] and deep ensembles [42].

VI. Intrinsically Interpretable Models

Intrinsically interpretable models are designed so that their decision-making processes can be understood directly. Examples include linear regression, logistic regression, decision trees, rule lists, and generalized additive models.

A. Linear and Logistic Regression

In a linear model $f(\mathbf{x}) = \beta_0 + \sum_i \beta_i x_i$, the contribution of each feature is represented by its coefficient: a unit increase in x_i changes the prediction by β_i, holding other features fixed. In logistic regression,

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + \exp(-(\beta_0 + \sum_i \beta_i x_i))}, \qquad (2)$$

and $\exp(\beta_i)$ is the multiplicative change in the odds for a unit increase in x_i. The interpretation is direct, but it depends on the assumption of linearity in the log-odds, and the coefficients can be unstable when features are strongly correlated. Sparsity-inducing regularization such as the Lasso helps by driving many coefficients to zero.

B. Decision Trees and Rule Lists

In a decision tree, predictions are obtained by following a sequence of understandable decision rules from the root to a leaf. A tree of small depth can be read, audited, and even printed on paper. Its interpretability decreases rapidly with depth: a tree with hundreds of leaves is no more transparent than a black-box model in practice. Rule lists and rule sets

TABLE I
QUALITATIVE COMPARISON OF MODEL FAMILIES (INDICATIVE)

Model family	Interpretability	Typical flexibility
Linear / logistic regression	High	Low
Shallow decision tree	High	Low
Generalized additive model	High–Medium	Medium
Random forest / boosted trees	Low	High (tabular)
Deep neural network	Low	Very high (unstructured)

provide a similar form of explanation, with an ordered list of if-then statements.

C. Generalized Additive Models

Generalized additive models have the form $g(\mathbb{E}[y]) = \beta_0 + \sum_i s_i(x_i)$, where each smooth function s_i can be plotted. Caruana *et al.* showed that GAMs with pairwise interactions achieve strong accuracy in healthcare while remaining intelligible, and that inspecting them revealed clinically surprising patterns, such as pneumonia patients with asthma being scored as lower risk because they had received more intensive care [43].

D. The Accuracy–Interpretability Trade-off

These models are often easier to interpret than deep neural networks. However, their simplicity may limit their ability to represent complex relationships in high-dimensional data such as images and raw text. The trade-off between interpretability and predictive performance depends on the task and dataset. For tabular data with meaningful features, the gap in accuracy between interpretable models and black boxes is often small, which motivates Rudin's argument that interpretable models should be preferred for high-stakes decisions [11]. For perceptual tasks, deep networks usually outperform simpler models by a wide margin, and post-hoc explanations are the main available option.

VII. POST-HOC EXPLANATION METHODS

Post-hoc methods explain predictions after a model has been trained. They are particularly useful when the underlying model is too complex to interpret directly.

A. Local Interpretable Model-Agnostic Explanations (LIME)

LIME explains an individual prediction by generating perturbed samples around an input instance and observing the predictions of the original model. It then fits a simple, interpretable surrogate model to approximate the original model's behavior near that instance [1].

Formally, LIME seeks an explanation $g \in G$ from a class G of interpretable models (for example, sparse linear models) by solving

$$\xi(\mathbf{x}) = \arg\min_{g \in G} \; \mathcal{L}(f, g, \pi_{\mathbf{x}}) + \Omega(g), \tag{3}$$

where $\pi_{\mathbf{x}}(\mathbf{z})$ is a proximity kernel that weights a perturbed sample $\mathbf{z}$ by its closeness to $\mathbf{x}$, $\Omega(g)$ penalizes the complexity of g, and the locality-aware loss is

$$\mathcal{L}(f, g, \pi_{\mathbf{x}}) = \sum_{\mathbf{z}, \mathbf{z}'} \pi_{\mathbf{x}}(\mathbf{z}) \big(f(\mathbf{z}) - g(\mathbf{z}')\big)^2. \tag{4}$$

Here $\mathbf{z}'$ is the interpretable representation of $\mathbf{z}$. For images, the interpretable representation is typically a binary vector indicating the presence or absence of contiguous regions called superpixels. For text, it indicates the presence or absence of words.

For example, LIME can identify which input features contributed most to a particular loan rejection prediction. Its explanation is local, meaning that it describes the model's behavior near the selected instance rather than the entire model. Because LIME relies on random sampling, explanations can vary between runs and with the kernel width.

B. SHapley Additive exPlanations (SHAP)

SHAP uses Shapley values from cooperative game theory to estimate the contribution of each feature to a prediction [2]. Features are treated as participants whose contributions are calculated based on their effects across different feature combinations.

Let $F = \{1, \dots, d\}$ be the set of features and let $v(S)$ denote the expected model output when only the features in $S \subseteq F$ are known. The Shapley value of feature i is

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(d - |S| - 1)!}{d!} \Big[v(S \cup \{i\}) - v(S) \Big]. \tag{5}$$

Intuitively, ϕ_i is the average marginal contribution of feature i over all possible orderings in which features can be added [22]. Shapley values are the unique attribution that satisfies the following properties: *efficiency* (the contributions sum to the difference between the prediction and the baseline, $\sum_i \phi_i = f(\mathbf{x}) - \phi_0$), *symmetry* (features that contribute equally receive equal values), *dummy* (a feature with no effect receives zero), and *additivity* (values for a sum of models are the sum of the values). SHAP casts the explanation as an additive feature-attribution model of the form in (1).

Exact computation of (5) requires 2^d evaluations, which is infeasible for large d. Practical variants include KernelSHAP, a model-agnostic estimator based on weighted linear regression; TreeSHAP, which computes exact values for tree ensembles in polynomial time [23]; and DeepSHAP, which combines Shapley values with the DeepLIFT propagation rules [24]. The definition of $v(S)$ also requires choosing how to treat absent features (marginal or conditional sampling), which matters when features are correlated.

SHAP provides explanations for individual predictions and can also support global analysis by aggregating feature contributions across a dataset, for example by averaging $|\phi_i|$ over instances to rank features or by plotting the distribution of ϕ_i against feature values. Its computational cost can vary considerably depending on the model and approximation method.

C. Gradient-weighted Class Activation Mapping (Grad-CAM)

Grad-CAM generates visual explanations for convolutional neural networks by using gradients flowing into a selected convolutional layer [3]. These gradients are used to produce a localization map that highlights image regions influential to a target class prediction.

Let $A^k \in \mathbb{R}^{u \times v}$ be the k-th feature map of the chosen convolutional layer and let y^c be the score for class c before the softmax. The importance weight of feature map k for class c is the global-average-pooled gradient

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \tag{6}$$

where $Z = u \cdot v$. The localization map is a weighted combination of the feature maps followed by a rectified linear unit,

$$L_{\text{Grad-CAM}}^c = \mathrm{ReLU}\left(\sum_k \alpha_k^c A^k \right). \tag{7}$$

The ReLU retains only features with a positive influence on the class of interest. The map has the spatial resolution of the chosen layer, so it is upsampled to the input size and normalized to $[0, 1]$ before being overlaid on the image, as in the second column of Fig. 2. Grad-CAM++ replaces the uniform weighting of gradients by a weighting derived from higher-order derivatives, which tends to improve localization when several instances of the same object appear [30].

A Grad-CAM heatmap supports investigation but does not establish that the model's reasoning is medically or otherwise correct.

D. Gradient-Based Attribution and Integrated Gradients

The simplest gradient-based attribution is the saliency map, $M_i = |\partial f(\mathbf{x}) / \partial x_i|$, which measures how sensitive the output is to a small change in each input element [25]. Integrated gradients address this issue by accumulating the gradient along a straight path from a baseline $\mathbf{x}'$ to the input $\mathbf{x}$ [29]:

$$\mathrm{IG}_i(\mathbf{x}) = (x_i - x_i') \int_0^1 \frac{\partial f\big(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}')\big)}{\partial x_i} \, d\alpha. \tag{8}$$

Integrated gradients satisfy completeness, meaning that the attributions sum to $f(\mathbf{x}) - f(\mathbf{x}')$, and sensitivity, meaning that a feature on which the output depends receives a non-zero attribution. The integral is approximated numerically, and the result depends on the chosen baseline.

E. Saliency Maps and Attention Visualization

Saliency maps highlight input regions that influence model predictions, particularly in computer vision. Attention visualization displays attention weights or related signals in models such as transformers [44], including applications in natural language processing.

These visualizations can provide useful insights into model behavior. However, attention weights and saliency values should not automatically be treated as complete or faithful explanations of a model's reasoning. Jain and Wallace showed that attention weights often correlate poorly with gradient-based importance [45], though Wiegreffe and Pinter argued the answer depends on how explanation is defined [46]. Adebayo *et al.* found that some saliency methods produce nearly the same maps when the trained weights of the network are replaced with random ones, which indicates that those maps depend more on the input than on the model [36]. Sanity checks should therefore precede reliance on any heatmap method.

F. Concept-Based and Example-Based Explanations

Concept activation vectors (CAVs) allow explanations at the level of human concepts. Given activations $h_l(\cdot)$ at layer l, a linear classifier is trained to separate activations of examples that contain a concept from activations of random examples, and its normal vector $\mathbf{v}_C^l$ is the CAV. The directional derivative of the class score $h_{l \to k}$ in the direction of the CAV is

$$S_{C,k,l}(\mathbf{x}) = \nabla h_{l,k}\big(f_l(\mathbf{x})\big) \cdot \mathbf{v}_C^l, \tag{9}$$

and the TCAV score is the fraction of examples of class k for which $S_{C,k,l} > 0$ [31]. A high TCAV score for the concept "striped" on the class "zebra" is the type of result illustrated in panel B of Fig. 1. Example-based methods instead explain a prediction by pointing to training instances. Influence functions estimate how the prediction would change if a training point were removed [32], and counterfactual explanations describe how an input would need to change to alter the outcome [17]. Counterfactuals are especially useful in lending or hiring, where the affected person wants to know what to change.

G. Feature-Effect Plots: PDP, ICE, and ALE

The partial dependence of the model on a feature subset S is the average prediction when the features in S are set to $\mathbf{x}_S$ and the other features C take their observed values:

$$\mathrm{PD}_S(\mathbf{x}_S) = \frac{1}{n} \sum_{k=1}^{n} f\big(\mathbf{x}_S, \mathbf{x}_C^{(k)}\big). \tag{10}$$

This is the quantity displayed in a PDP [33]. An ICE plot shows the same curve for each instance separately and so reveals heterogeneity that the average can hide [34]. PDPs assume that the features in S are independent of those in C; when features are correlated, they average over combinations that are unrealistic. ALE plots address this by averaging local changes in the prediction over a conditional distribution rather than the marginal [35]. Panel C of Fig. 1 sketches the three plot types.

VIII. Comparison of Major Methods

A. Qualitative Comparison on Image Data

Fig. 2 shows how Grad-CAM, LIME, and SHAP explain the predictions of an image classifier on four examples. Each row presents the original image followed by the explanation from each method. The Grad-CAM column is a continuous heatmap scaled from 0 to 1, the LIME column outlines the superpixels that were most supportive of the predicted class, and the SHAP

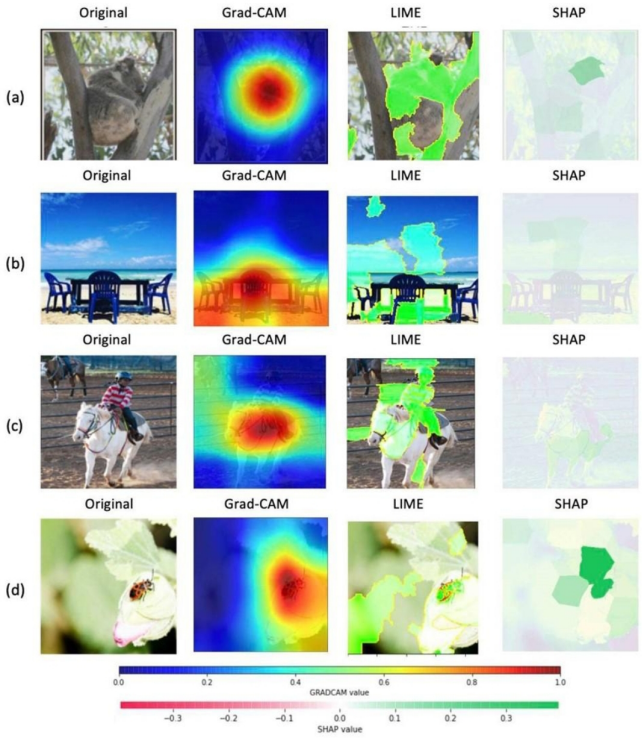


Fig. 2. Qualitative comparison of Grad-CAM, LIME, and SHAP explanations for four example images, (a)–(d). Grad-CAM values range from 0 to 1, and SHAP values are shown on a signed scale. Adapted from the literature; source to be cited.

column shades superpixels using Shapley values, with green indicating a positive contribution and red a negative one.

The three methods do not always agree. In (a), Grad-CAM concentrates on a compact region near the animal's body, LIME marks irregular regions including surrounding branches, and SHAP assigns a positive value to a single superpixel. In (b), SHAP values are near zero almost everywhere, so no single superpixel has a strong effect. In (d), where a small insect occupies part of the frame, all three concentrate on the insect and agree far more. The methods answer different questions: Grad-CAM shows where final convolutional features supporting the class lie, LIME shows which regions change the prediction in a local linear model, and SHAP gives average marginal contributions of regions. Grad-CAM is coarse because of layer resolution, while LIME and SHAP depend on the superpixel segmentation. Highlighted background may reflect genuine contextual cues or an imperfect explainer, which the figure alone cannot resolve; this motivates the evaluation in Section IX.

B. Summary of Method Properties

Table II summarizes the main characteristics of the methods described above.

No single method is best. TreeSHAP is a strong default for tree ensembles, Grad-CAM gives a fast check for convolutional models, and LIME or KernelSHAP serve query-only black boxes. Counterfactual explanations suit non-expert audiences.

IX. Evaluation of Explanations

Doshi-Velez and Kim proposed a three-level taxonomy of evaluation [6].

A. Application-Grounded Evaluation

The explanation is evaluated within the real task by domain experts, for example, by measuring whether radiologists make better decisions with the help of a heatmap. This is the most convincing form of evaluation but also the most costly.

B. Human-Grounded Evaluation

Simplified tasks are given to non-expert participants, for instance, asking which of two explanations is clearer or whether the participant can predict the model's output given the explanation. Controlled studies show that transparency does not always help people detect model errors [47].

C. Functionally Grounded Evaluation

Proxy metrics are used without human subjects. Common functional criteria include:

- **Fidelity (faithfulness):** how well the explanation agrees with the model. For attributions, a common test is *deletion*: remove or mask the features with the highest attribution and measure the drop in the predicted score; a faithful method should produce a rapid drop. A complementary *insertion* test adds the most important features to a blank input and measures the rise.
- **Stability (robustness):** a local Lipschitz-style estimate such as

$$\hat{L}(\mathbf{x}) = \max_{\mathbf{x}' \in N_\epsilon(\mathbf{x})} \frac{\|e(\mathbf{x}) - e(\mathbf{x}')\|}{\|\mathbf{x} - \mathbf{x}'\|} \tag{11}$$

  quantifies how much the explanation can change for a small change of input [39].
- **Sparsity:** the number of features with non-negligible attribution.
- **Model-parameter randomization:** the explanation should change when the model's learned weights are randomized [36].
- **Remove-and-retrain:** features ranked as important are removed and the model is retrained to see whether performance falls more than for random removal, which controls for the distribution shift caused by simple masking [48].

Kindermans *et al.* showed some saliency methods change under input shifts that do not affect predictions [49], so a plausible-looking explanation is not evidence of correctness.

TABLE II
COMPARISON OF REPRESENTATIVE XAI METHODS

Method	Scope	Model access	Data type	Main strength	Main limitation
LIME	Local	Agnostic	Tabular, text, image	Simple, flexible, intuitive	Sampling variance; sensitive to kernel and segmentation
SHAP	Local and global	Agnostic or specific (TreeSHAP)	Mostly tabular; also image and text	Axiomatic basis; consistent attributions	Cost grows with features; depends on treatment of absent features
Grad-CAM	Local	Specific (CNN)	Image	Fast; class-discriminative heatmap	Coarse resolution; needs gradient access
Saliency / gradients	Local	Specific	Image, text	Very fast	Noisy; saturation; can fail sanity checks
Integrated gradients	Local	Specific	Image, text, tabular	Completeness axiom	Baseline-dependent
TCAV	Global (per class)	Specific	Image mostly	Human-level concepts	Needs labeled concept examples
PDP / ICE / ALE	Global / local	Agnostic	Tabular	Shows effect shape	PDP assumes independence; low-dimensional only
Decision tree / GAM	Global	Intrinsic	Tabular	Transparent by design	Limited capacity on raw data

X. APPLICATIONS

A. Healthcare

XAI can help clinicians inspect predictions from imaging systems and models trained on health records. Deep networks reach expert-level performance on skin-lesion and chest-radiograph tasks [13], [14], and heatmaps or feature attributions can support review [50]. Explanations should not replace professional judgment: Ghassemi *et al.* caution that current post-hoc explanations are not reliable at the level of the individual patient and do not substitute for validation [51]. In the pneumonia study of Caruana *et al.*, an interpretable model exposed a counterintuitive asthma rule, showing how transparency can prevent harm [43].

B. Finance

Institutions use XAI to investigate credit scoring, fraud detection, and loan approval. Local attributions or counterfactuals can communicate the principal reasons for a refusal, support internal audits, and flag inappropriate behavior. SHAP on gradient-boosted trees is popular because it is fast and its values sum to the model output.

C. Autonomous Systems

In vehicles and robots, explanations help engineers investigate perception and decision modules, for example whether a detector attends to pedestrians and lane markings or to irrelevant structures, and support post-incident analysis.

D. Criminal Justice

XAI can support auditing of risk-assessment systems for influential factors and group disparities. Because decisions affect individual rights, explanations are not proof of fairness, and Rudin argues that inherently interpretable models are preferable in such settings [11].

E. Natural Language Processing

Text explanations take the form of word-level scores, attention maps, or generated rationales, and can reveal reliance on spurious tokens, though attention alone is not a reliable explanation [45], [46].

XI. REGULATORY, LEGAL, AND ETHICAL DIMENSIONS

A. Regulatory Context

Data protection and AI governance frameworks have increasingly emphasized transparency. Discussion of the European Union's General Data Protection Regulation (GDPR) drew attention to provisions on automated decision-making and the information that data subjects can receive about the logic involved, and scholars have disagreed about whether these provisions amount to a legally binding "right to explanation" [16], [17]. More recent AI-specific regulation in several jurisdictions introduces requirements on transparency, documentation, and human oversight for systems classified as high-risk. The precise obligations differ by jurisdiction, sector, and risk category and continue to evolve, so organizations should consult current legal texts and qualified counsel rather than rely on a general survey.

B. Ethical Considerations

Explanations raise ethical questions of their own. An explanation can be *persuasive* without being *faithful*, and a user may place more trust in a system because of a convincing but misleading explanation. This risk is sometimes called explanation-based "fairwashing" when explanations are used

to make a discriminatory model look acceptable [38]. Explanations may also expose proprietary models or enable gaming of the system by individuals who learn exactly which factors are decisive. Finally, explanation interfaces should respect privacy, since feature-level explanations on sensitive data could reveal personal information.

C. Documentation Practices

Explanations are most useful when they are part of a broader documentation and governance practice that includes dataset documentation, model cards describing intended use and limitations, performance reporting across subgroups, monitoring after deployment, and procedures for human review and appeal. XAI methods supply evidence for these processes but do not replace them.

XII. Challenges and Limitations

A. Fidelity Versus Interpretability

Simple explanations are easier to understand but may misrepresent complex models, while detailed ones overwhelm non-experts. A local linear surrogate may be accurate near an instance and poor outside it, and users may extrapolate incorrectly.

B. Stability

Small input changes can produce very different explanations even when the prediction is unchanged. Ghorbani *et al.* crafted imperceptible perturbations that preserve the label but alter the interpretation [37], and Alvarez-Melis and Jaakkola found popular methods lack robustness [39].

C. Evaluation Metrics

No universal metric captures every aspect of explanation quality, and masking-based metrics can be confounded by distribution shift [48].

D. Human Factors

Engineers may need technical contributions, whereas patients and customers need concise plain-language explanations. People prefer contrastive, selective, cause-based explanations [10]. Attributions show what the model uses, not what causes the outcome in the world.

E. Computational Cost

Exact SHAP needs exponentially many evaluations in the number of features, and LIME needs thousands of queries per instance. Approximations reduce cost but add uncertainty.

F. Adversarial Manipulation

Slack *et al.* built classifiers that behave discriminatorily on real data but innocuously on the perturbed samples used by LIME and SHAP, hiding bias from the explainer [38]. Explanation tools should not be trusted as audit instruments without safeguards.

G. Correlated Features and Foundation Models

With correlated features, credit for an effect can be split in many ways. Large foundation models, with billions of parameters and open-ended outputs, strain methods designed for single-label classifiers, and evaluation standards remain immature.

XIII. Future Research Directions

Key directions include:

- **Standardized benchmarks:** combining planted-shortcut tasks, known ground truth, and human studies [6], [48].
- **Robust, faithful explanations:** stable under perturbation, resistant to manipulation, and reporting their own uncertainty.
- **Human-centered interfaces:** interactive, contrastive, layered explanations evaluated by their effect on decision quality and calibrated trust [10], [47].
- **Causal and concept-level explanations:** moving beyond pixel and feature attributions [31].
- **Interpretable-by-design models:** modern GAMs, prototype networks, concept bottleneck models, and rule sets [11], [43].
- **Foundation models and governance:** scalable explanations for generative models, integrated with testing, documentation, and auditing.

XIV. Conclusion

Explainable Artificial Intelligence is an important component of developing trustworthy, accountable, and fair AI systems, particularly in critical domains such as healthcare and finance. This paper reviewed the need for explainability, a taxonomy of methods, intrinsically interpretable models, and post-hoc techniques such as LIME, SHAP, and Grad-CAM with their mathematical formulations and a qualitative comparison on image data. It also discussed evaluation, applications, regulation, and challenges.

Future research should focus on establishing standardized evaluation benchmarks, improving the robustness of explanations against adversarial manipulation, and designing human-centered interfaces that communicate model behavior effectively to different user groups. Further work is also needed to assess whether explanations genuinely improve human decision-making and help identify harmful model behavior.

As AI systems become more widely deployed, explainability will continue to play an important role in responsible AI development. However, explanations should be evaluated carefully and combined with appropriate testing, auditing, and human oversight rather than treated as guarantees of correctness or fairness.

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